

on the impurity via an OTOC given by

$$B(v_d, T) = \frac{1}{4} \text{tr} \left\{ \rho_0^{IM} [\tilde{X}^{(I)}(T), Y^{(I)}]^\dagger [\tilde{X}^{(I)}(T), Y^{(I)}] \right\}, \quad (4)$$

whose definition is equivalent to that of $C^{(I)}(T)$, but with the operators evolved under the backflow protocol, i.e. $\tilde{X}^{(I)}(T) = U_2^\dagger U_1^\dagger X^{(I)} U_1 U_2$. We enforce that $t_1, t_2 \gg p^{-1}$ such that the impurity and medium have significant time to interact before the reset step and subsequent OTOC readout.

In the case of random unitary evolution the environment strongly scrambles any information that flows into it and the system evolution is Markovian as characterized by a process tensor which factorizes between timesteps. However, it has recently been shown that although the state evolution may be Markovian or near-Markovian, the OTOC can still display non-Markovian correlations [89]. These correlations arise due to the Heisenberg evolution of the operator; in our model this causes the support of the operator to first spread onto the medium and then, after the reset step, onto the fresh impurity. This is highlighted in Figure 2 where we illustrate how information of the initial state can flow through the various channels of the process. The only way for this information to flow onto the final impurity is via the channel represented by the blue line. This corresponds to backflow of information from the medium via U_2 , this information having flowed into the medium via U_1 . We expect U_1 to transfer information from the impurity to the medium regardless of the impurity's velocity, but U_2 can only implement the reverse if the impurity remains causally connected to the initial transfer, i.e. if it travels at sub-luminal velocities. Although we treat the full system unitarily, by considering the impurity sector as the system and the medium as the environment, we can consistently refer to the Markovianity of the reduced operator dynamics on the impurity.

4 Operator dynamics

In this section, we present the OTOC dynamics for both the 1D and 2D models introduced in Section 2. First, we investigate the dynamics of the backflow protocol for the 1D system and perform finite-time scaling on the late-time dynamics. For the 2D system, we consider the steady state of the OTOC without applying the reset step and investigate the effect of operator backflow on the scrambling of operators within the 1D impurity.

4.1 One-dimensional medium

We shall focus exclusively on $\overline{B(v_d, T)}$, the disorder-averaged weight of the OTOC on the impurity at time T during the reset protocol depicted in Figure 4. In Figure 5 we plot $\overline{B(v_d, T)}$ at a late time as a function of the drift velocity v_d . A transition occurs between at a critical velocity of $v^* \approx 1.2$. To understand the significance of this value we must recall that the information dynamics within the medium is stochastic. So, while the rate at which the lightcone broadens is bounded from above by c , we can also associate a butterfly velocity, v_B , with the typical expansion rate. The average increase in the number of sites on which an operator has support after a time t is then given by $2v_B t$ [68]. In our case we have $v_B = 1.2$, precisely the velocity at which we observe the transition.

In Figure 6 we show that the disorder-averaged backflow satisfies a finite-time scaling of the following form:

$$\overline{B(v_d, T)} = f((v_d - v^*)\sqrt{T}), \quad (5)$$

where f is a universal scaling function. The origin of this exponent can be understood by analyzing the operator dynamics within the medium. Given that t_2 is much larger than